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肝毒性门户 (HTP): 一个整合的药物诱导肝毒性知识库和基于图神经网络的预测模型数据库

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摘要

肝毒性在药物开发中构成了关键挑战, 这是由于肝脏在药物代谢和解毒中的核心作用。准确预测肝毒性至关重要, 但受到信息来源分散、缺乏整理标准以及数据视角异质性的阻碍。为了解决这些挑战, 我们开发了肝毒性门户 (HepatoToxicity Portal, HTP), 该门户整合了专家整理的知识库 (HTP-KB) 和用于毒性预测的最先进机器学习模型 (HTP-Pred)。HTP-KB 汇总了来自九个主要数据库的肝毒性数据, 由肝毒性专家仔细审查, 并使用《医药监管活动词典》(MedDRA) 术语分为三类: 体外、体内和临床。该知识库涵盖 8,306 种化学物质的信息。此整理数据集用于建立肝毒性预测模块, 通过微调基于图神经网络 (GNN) 的基础模型, 该模型预训练了大约

科学贡献

HTP-KB 汇总了来自九个来源的关于肝脏毒性的全面整理信息。HTP-Pred 利用先进的深度学习技术, 显著提高预测准确性。这些工具共同为药物开发的研究人员和从业者提供了宝贵的资源, 可通过用户友好的界面访问。

† 韩智妍、郑元昊和张仁洙对本工作贡献相同。

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DATABASE

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HepatoToxicity Portal (HTP): an integrated database of drug-induced hepatotoxicity knowledgebase and graph neural network-based prediction model

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Abstract

Liver toxicity poses a critical challenge in drug development due to the liver's pivotal role in drug metabolism and detoxification. Accurately predicting liver toxicity is crucial but is hindered by scattered information sources, a lack of curation standards, and the heterogeneity of data perspectives. To address these challenges, we developed the HepatoToxicity Portal (HTP), which integrates an expert-curated knowledgebase (HTP-KB) and a state-of-the-art machine learning model for toxicity prediction (HTP-Pred). The HTP-KB consolidates hepatotoxicity data from nine major databases, carefully reviewed by hepatotoxicity experts and categorized into three levels: in vitro, in vivo, and clinical, using the Medical Dictionary for Regulatory Activities (MedDRA) terminology. The knowledgebase includes information on 8,306 chemicals. This curated dataset was used to build a hepatotoxicity prediction module by fine-tuning a GNN-based foundation model, which was pre-trained with approximately 10 million chemicals in the PubChem database. Our model demonstrated excellent performance, achieving an area under the ROC curve (AUROC) of 0.761, surpassing existing methods for hepatotoxicity prediction. The HTP is publicly accessible at <https://kobic.re.kr/http/>, offering both curated data and prediction services through an intuitive interface, thus effectively supporting drug development efforts.

Scientific contributions

HTP-KB consolidates comprehensive curated information on liver toxicity gathered from nine sources. HTP-Pred utilizes advanced deep learning techniques, significantly enhancing predictive accuracy. Together, these tools provide valuable resources for researchers and practitioners in drug development, accessible through a user-friendly interface.

† Jiyeon Han, Wonho Zhung and Insoo Jang have contributed equally to the work.

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关键词 肝毒性, 药物性肝损伤, 肝毒性分子图, 微调, 基础模型, 深度神经网络, 图神经网络

介绍

药物开发是一个复杂且资源密集的过程, 每个开发阶段的成功率不到10% [1, 2]。导致这种高淘汰率的一个重要因素是药物毒性, 而动物模型与人体反应之间的差异常常会加剧这种毒性 [3–6]。鉴于肝脏在化学转化和解毒中的关键作用, 它尤其容易受到药物引起的损伤。即使在FDA批准上市后, 药物仍可能产生不良反应, 例如药物性肝损伤 (DILI), 这是美国三级护理中心急性肝衰竭病例的主要原因, 占病例的50%以上 [7]。

对于详细描述药物对肝组织影响的综合知识库的需求已经变得明显。美国FDA已经做出了重大努力, 以建立针对FDA批准药物的DILI知识资源。肝脏毒性知识库 (Liver Toxicity Knowledge Base, LTKB) 是一个总括性项目, 旨在开发内容丰富的肝脏毒性资源 [8]。值得注意的是, DILIrank数据集 [9] 是将1,036种FDA批准药物根据其引起DILI的潜在性分为四类的分类系统, 这一分类是通过分析药物标签文件中的肝毒性描述并评估文献中的因果证据确定的。类似地, LiverTox [10] 为1,400多种药物提供了关于DILI的临床和研究信息。这些数据库在肝毒性研究中具有关键作用, 但其覆盖范围仅限于市场上的药物。

实验数据仍然至关重要, 因为它提供了药物在细胞和生物体水平上作用的详细见解。像 InvitroDB [11] 和 CEBS [12] 这样的数据库体现了基于细胞系药物实验对生物系统中化学物质作用进行分类的努力, 尽管将这些发现转化为临床见解仍然是一个挑战。其他方法包括汇编来自多篇出版物的药物实验结果, 以提供关于药物作用的多样化视角 [13–15]。然而, 这些数据库的可用性通常受到其参考数据格式的限制, 这些数据通常以 PDF 或 CSV 格式存储, 从而增加了研究人员提取数据的难度。

为了方便访问全面的药物数据, 各种网络服务器已被开发以整合不同的资源。例子包括 CompTox [16]、NITE-CHIRP [17], 和 eChemPortal [18], , 分别提供基于网络的访问美国、日本和经合组织的毒性参考数据库。然而, 评估整体化合物毒性或发现隐藏的

生物学连接仍然具有挑战性, 因为这些平台通常缺乏额外的策划和数据可视化功能。

最近的研究集中于利用汇编数据集开发肝毒性预测模型, 这些数据集反映了多样的生物学情景。与传统的体外和体内实验相比, 计算方法在时间、覆盖范围和成本效率方面具有优势。Greene 等人提出了一种利用 ECFP6 指纹分类预定义肝毒性标签的模型, 为后续的算法进展铺平了道路。贝叶斯模型、支持向量机 (SVM)、决策树和随机森林自此被广泛应用于预测肝毒性, 通常结合集成方法以提高预测性能。

随着深度学习方法的出现, 基于卷积神经网络 (CNN) 的方法也被用于毒性预测。Kang 等人应用深度神经网络来表示化合物指纹, 以进行肝毒性预测 [33], 而 Xu 等人则利用无向图递归神经网络对分子结构进行编码, 以识别 DILI 阳性分子 [34]。这些方法展示了深度学习在将化学结构和性质与肝毒性结果联系起来方面的潜力, 值得进一步探索先进的算法和方法。

除了算法研究之外, 还努力提供用户友好的网页服务器, 提供预测模型和毒性数据。PASS Online 支持多种基于文献数据训练的预测模块, 并进行积极维护 [35]。同样地, LAZAR [36]、ProTox 3 [37]、admetSAR 2.0 [38], 和 eMolTox [39] 提供专注于 ADMET (吸收、分布、代谢、排泄和毒性) 各方面的预测模块。然而, 这些平台在主题覆盖方面虽然全面, 但在针对肝毒性进行深入分析方面仍显不足。

为了响应这些需求, 我们推出了肝毒性门户 (HTP), 这是一个专注于肝脏毒性的专业网络应用。HTP整合了来自多种毒性数据库的精选数据, 并呈现了基于大规模数据集训练的准确预测模型。我们的知识库系统地根据多个参考来源分类整理了肝毒性化合物, 汇编了通过人工整理的肝毒性评分, 这对非毒理学和毒理学领域都可能具有价值。

Keywords Liver toxicity, Drug induced liver injury, Hepatotoxicity, Molecular graph, Fine-tuning, Foundation model, Deep neural network, Graph neural network

Introduction

Drug development is a complex and resource-intensive process with a low success rate of less than 10% in each developmental phase [1, 2]. A significant contributing factor to this high attrition rate is drug toxicity, often exacerbated by discrepancies between animal models and human responses [3–6]. Given the liver’s pivotal role in chemical transformation and detoxification, it is particularly susceptible to drug-induced damage. Even after FDA market approval, drugs may have adverse effects such as drug-induced liver injury (DILI), a major cause of acute liver failure cases in U.S. tertiary care centers, accounting for over 50% of instances [7].

The need for comprehensive knowledge bases detailing drug effects on liver tissues has become apparent. The US FDA has made significant efforts to establish knowledge resources of DILI for FDA-approved drugs. The Liver Toxicity Knowledge Base (LTKB) is an umbrella project to develop content-rich resources on liver toxicity [8]. Notably, the DILIrank dataset [9] is the classification of 1,036 FDA-approved drugs into four classes according to their potential for causing DILI, determined by analyzing the hepatotoxic descriptions in the drug labeling documents and assessing causality evidence in literature. Similarly, LiverTox [10] provides clinical and research information on DILI for over 1,400 drugs. These databases are pivotal in hepatotoxicity research, yet their coverage is limited to drugs in the market only.

Experimental data remains crucial as it offers detailed insights into drug effects at cellular and organismal levels. Databases like InvitroDB [11] and CEBS [12] exemplify efforts to catalog chemical effects in biological systems based on drug experiments in cell lines, though translating these findings into clinical insights remains a challenge. Other approaches involve compiling drug experimental results from multiple publications to offer diverse perspectives on drug effects [13–15]. However, the usability of these databases is often hindered by the format of their reference data, typically stored as PDFs or CSVs, complicating data extraction for researchers.

To facilitate access to comprehensive drug data, various web servers have been developed to integrate disparate resources. Examples include CompTox [16], NITE-CHIRP [17], and eChemPortal [18], providing web-based access to toxicity reference databases in the U.S., Japan, and OECD, respectively. However, assessing overall compound toxicity or uncovering hidden

biological connections remains challenging, as these platforms often lack additional curation and data visualization features.

Recent studies have focused on developing predictive models for hepatotoxicity using compiled datasets, reflecting diverse biological scenarios. Computational methods offer advantages over traditional in vitro and in vivo experiments in terms of time, coverage, and cost efficiency. Greene et al. introduced a model utilizing ECFP6 fingerprints to classify predefined hepatotoxicity labels [19], paving the way for subsequent algorithmic advancements. Bayesian models [20, 21], support vector machines (SVMs) [22–24], decision trees [25, 26], and random forests [24, 27, 28] have since been widely applied to predict hepatotoxicity, often integrating ensemble methods to enhance predictive performance [29–32].

With the emergence of deep learning methods, convolutional neural network (CNN)-based approaches have also been employed for toxicity predictions. Kang et al. applied deep neural networks to represent fingerprints of chemical compounds for hepatotoxicity prediction [33], while Xu et al. utilized undirected graph recursive neural networks for molecular structure encoding to identify DILI-positive molecules [34]. These approaches demonstrate the potential of deep learning in linking chemical structures and properties with hepatotoxicity outcomes, warranting further exploration of advanced algorithms and methodologies.

Beyond algorithmic research, efforts have been made to provide user-friendly web servers offering both prediction models and toxicity data. PASS Online supports diverse prediction modules trained on literature data with active maintenance [35]. Similarly, LAZAR [36], ProTox3 [37], admetSAR 2.0 [38], and eMolTox [39] provide prediction modules focusing on various aspects of ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity). However, these platforms, while comprehensive in terms of subject coverage, lack in-depth analysis specific to hepatotoxicity.

In response to these needs, we introduce the HepatoToxicity Portal (HTP), a specialized web application focused on liver toxicity. HTP integrates curated data from diverse toxicity databases and presents accurate prediction models trained on extensive datasets. Our knowledgebase systematically catalogs hepatotoxic compounds based on multiple reference sources, compiling the hepatotoxicity scores with manual curation, which could be valuable for both non-toxicology and toxicology

研究人员。此外, 为了应对基于生物学的深度学习模型中持续存在的数据稀缺问题, HTP 利用了一种通用预训练的分子图模型以及微调技术, 从而比现有方法提高了性能。

结构与内容

数据库概览

HTP由两个模块组成, 即HTP知识库 (HTP-KB) 和 HTP预测 (HTP-Pred) (图1)。HTP-KB作为一个知识库, 整合了来自九个公共资源的信息。在从 PubChem标注化合物ID [40], 后, 收集的文献经过人工整理, 将其信息分类为三类: 临床、体内和体外证据。此外, 根据每种化合物预期的生物学机制, 对《医学监管活动词典》(MedDRA) 中的肝毒性相关术语进行了标注。综合肝毒性评分的计算考虑了每个参考文献内容的方法学重要性。随后的模块HTP-Pred是一种肝毒性预测工具, 利用在大规模未标记分子数据上预训练的图神经网络, 并使用我们整理的文献进行微调。

用于肝毒性预测的数据集。HTP-KB 和 HTP-Pred 已整合到一个具有增强可视化的网页信息门户中。用户可以预测新小分子的毒性评分并识别亚结构毒性基团。

数据收集与整理

数据收集与整合

HTP-KB 包含一个完整的九个化学相关数据库的集合, 每个数据库都是为不同的目标和隶属关系而建立的 (表 1)。这些数据库根据其特定用途进行分类, 包括整理药物实验结果 (CEBS [12], InvitroDB [11])、汇总市售药物信息 (DrugBank [41], DILIrank [9], SIDER [42], LiverTox [10]) 以及整理有关药物-环境效应的案例研究及相关出版物 (T3DB [13], IRIS [14], ATSDR [15])。

根据数据库的不同, 肝脏特异性内容要么可以直接获取, 要么需要从完整数据集中进行额外筛选。下载的数据集经过人工筛选, 以确保其与肝脏毒性相关。在整个注释过程中, 采用了在大多数数据库中广泛使用的 PubChem化合物标识符 (CID)。在需要分配的情况下

researchers. Moreover, to address the persistent issue of data scarcity in biology-based deep learning models, HTP leverages a generally pre-trained molecular graph-based model and fine-tuning techniques, resulting in improved performance compared to existing methods.

Construction and content

Database overview

The HTP comprises two modules, namely the HTP KnowledgeBase (HTP-KB) and HTP Prediction (HTP-Pred) (Fig. 1). HTP-KB serves as a knowledgebase, consolidating information from nine public resources. After annotating the compound ID from PubChem [40], the collected documents underwent manual curation classifying their information into three classes: clinical, in vivo, and in vitro evidence. Additionally, liver toxicity-related terms from Medical Dictionary for Regulatory Activities (MedDRA) were annotated based on the anticipated biological mechanisms of each compound. The overall hepatotoxicity score was computed considering the methodological importance of each reference of the contents. The subsequent module, HTP-Pred, is a hepatotoxicity prediction tool that leverages a pre-trained graph neural network on large unlabeled molecule data, which is fine-tuned using our curated

dataset for hepatotoxicity prediction. HTP-KB and HTP-Pred are integrated into a web information portal with enhanced visualizations. Users can predict the toxicity score of new small molecules and identify substructural toxicophores.

Data collection and curation

Data collection and integration

The HTP-KB comprises a comprehensive collection of nine chemical-related databases, each established with diverse objectives and affiliations (Table 1). These databases are categorized based on their specific purposes, including organizing results from drug experiments (CEBS [12], InvitroDB [11]), aggregating information on commercially available drugs (DrugBank [41], DILIrank [9], SIDER [42], LiverTox [10]), and curating case studies on drug-environment effects along with relevant publications (T3DB [13], IRIS [14], ATSDR [15]).

Depending on the database, liver-specific content was either readily accessible or required additional filtering from the complete dataset. The downloaded dataset underwent manual filtering to ensure its relevance to liver toxicity. Throughout the annotation process, PubChem Compound Identifiers (CIDs), widely used across most databases, were employed. In cases where assigning

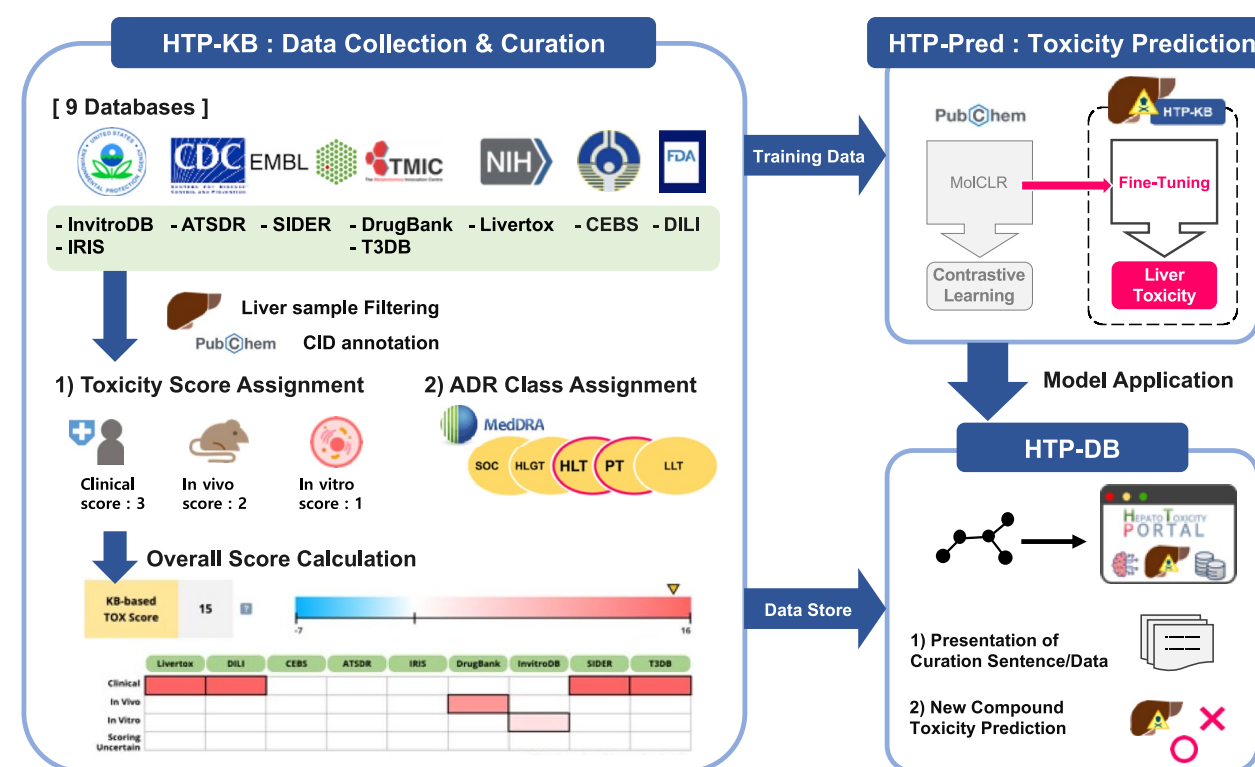


图 1A HTP 数据库概述, 说明了知识库、预测模块和网页门户的发展

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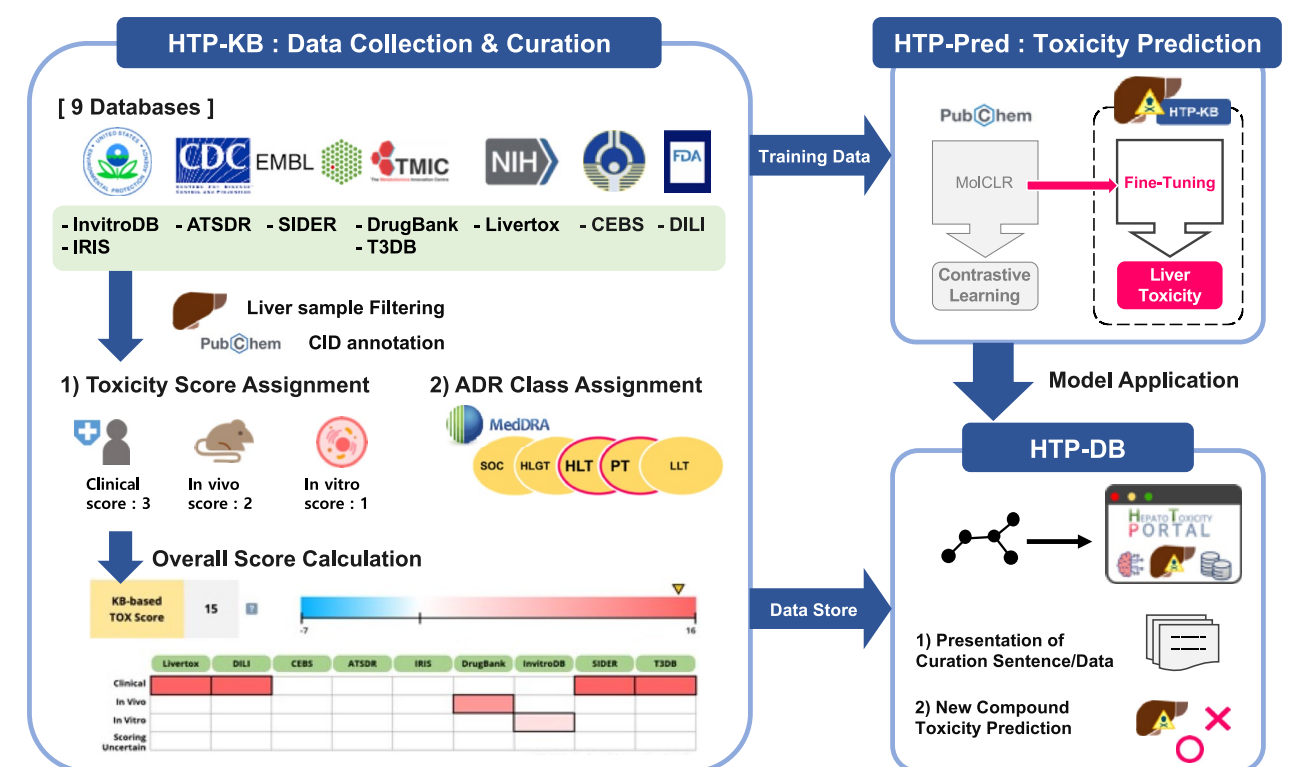


Fig. 1 An overview of the HTP database, illustrating the development of the knowledgebase, prediction module, and web portal

Table 1 Collection and characteristics of databases for HTP-KB

Database (Download date)	Affiliations	Hepatotoxicity information accessibility	Toxicity classification	Data format	No. of compound entries	No. of final curated CIDs	Data class
InvitroDB v3.4 (2022.02.04)	EPA (U.S.)	Low	Δ	CSV	9329	6466	In vitro
T3DB (2022.07.20)	TMIC (Canada)	Low	O	CSV	3673	955	In vitro In vivo clinical
LiverTox (2022.01.27)	NIH (U.S.)	High	O	PDF	1133	1005	In vitro In vivo clinical
DILIrank (2022.01.27)	FDA (U.S.)	High	O	PDF	1036	669	Clinical
SIDER (2022.02.04)	EMBL (Europe)	Middle	O	PDF	1430	749	Clinical
IRIS (2022.01.28)	EPA (U.S.)	Middle	Δ	PDF	666	138	In vivo clinical
CEBS (2022.02.04)	NTP (U.S.)	Low	X	CSV	9209 CASID	230	In vivo
ATSDR (2022.02.03)	CDC (U.S.)	Low	X	PDF	216	145	In vivo clinical
DrugBank v5.1.7 (2020.09.08)	TMIC (Canada)	Low	X	XML	13,580	154	In vitro In vivo clinical

* Hepatotoxicity accessibility levels (High, Easily accessible to liver-related toxicity results; Middle, Liver-related toxicity data available, yet needs additional filtration; Low, Difficult to extract liver-related results), Toxicity classification levels (O, Classification system is easily interpreted; Δ, Classification exists, yet not systematic; X, There is no classification levels)

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使用了PubChemPy (版本1.0.4), 这是一个利用各种物质标识符检索相关化合物数据的工具, 因为一个独特的PubChem CID不明确。由于各数据库可用数据的差异, 详细的整理过程有所不同(补充图S1)。在过滤前后的具体数据量在补充材料中列出。

MedDRA 注释

为了使用标准化词汇描述生物学活动, 我们利用了《法规事务医学词典》(MedDRA) 术语 [43] 来注释从九个数据库汇总的文献和参考资料。MedDRA 是一个国际医学本体, 支持范围广泛的制药和医学主题, 其结构分为系统器官分类(SOC)下的四个层级(补充图S2)。MedDRA 本体可通过 BioPortal 访问(BioPortal MedDRA 2019AB, 访问时间 2019.11.18)。MedDRA 结构的四个层级包括高层组术语(HLGT)、高层术语(HLT)、优选术语(PT)和最低层术语(LLT)。在注释与肝毒性相关的参考资料时, 我们重点关注 SOC 级别的术语“肝胆系统疾病”和“检查”, 并从 BioPortal 提取其子层级数据。在“肝胆系统疾病”下, 我们选择了五个 HLGT 术语: “肝及肝胆系统疾病”、“肝胆肿瘤”、

化合物(c)的评分是根据来自九个来源数据库的所有记录的贡献加权总和计算的, 同时考虑每条记录对肝毒性的正面或负面影响:

$$S_c = \sum_{i=1}^{n_c} \text{sign}_c(i) \times \text{weight}_c(i) \quad (1)$$

其中: n_c , 化合物c的记录数量; $\text{sign}_c(i)$, +1 或−1, 取决于记录描述的是肝毒性阳性还是阴性证据; $\text{weight}_c(i)$, 分别为临床、体内或体外证据的权重, 分别为3、2或1。

数据库中的冲突记录不计入总和(即权重为零)。该评分系统允许我们根据来自不同来源的汇总证据评估化合物的整体肝毒性潜力, 同时考虑每个数据库提供的数据质量和类型的差异。

HTP - KB 内容和统计

通过整合九个数据库, 随后进行人工整理和评分, 已创建了最全面的肝毒性知识库。我们在图 2 中简要概述了 HTP-KB 内容的统计信息, 包括证据类别、来源数据库、注释水平以及总体肝毒性评分。此外, 每个数据库的详细贡献和化合物重叠情况在补充图 S3 中展示。所有统计数据均基于 PubChem CID。

HTP-KB 包含总计 8306 种化合物, 由毒理学专家手工整理为三类。其中有 2260 条 (27.2%) 条目有临床证据支持, 显著超过 LiverTox 或 DILI-Irank 中的条目(图 2a 和 b)。由体外证据支持的条目占大部分, 有 6472 种 (77.9%) 化合物, 这表明 HTP-KB 通过整合体外证据显著扩展了肝毒性化合物的范围。

分析记录的源数据库, 临床类别中的2260条目分布在LiverTox (1005)、T3DB (890)、SIDER (748)和DILI (669)等数据库中(图2b)。CEBS提供了大量的体内证据, 但在知识库中所占比例较小。几乎所有的体外证据都来源于InvitroDB。

接下来, 我们检查了数据库中肝毒性评分的分布, 范围从 − 7 到 +16 (图 2c)。直方图显示分布偏向正面, 这可能是因为通常相比负肝毒性, 更容易确定正肝毒性。

a unique PubChem CID was unclear, PubChemPy (ver.1.0.4), a tool for retrieving related compound data using various substance identifiers, was utilized. The detailed curation processes varied due to disparities in available data across databases (Supplementary Fig. S1). The specific quantities of data before and after filtering is outlined in the Supplementary Materials.

MedDRA annotation

To describe biological activities with standardized vocabularies, we utilized the Medical Dictionary for Regulatory Activities (MedDRA) terms [43] to annotate documents and references aggregated from nine databases. MedDRA is an international medical ontology that supports a wide range of pharmaceutical and medical subject structured into four hierarchical levels under the System Organ Class (SOC) (Supplementary Fig. S2). The MedDRA ontology was accessed via BioPortal (BioPortal MedDRA 2019AB, accessed 2019.11.18). The four levels of the MedDRA structure consist of High-Level Group Terms (HLGT), High-Level Terms (HLT), Preferred Terms (PT), and Lowest Level Terms (LLT). For annotating references related to hepatotoxicity, we focused on SOC-level terms ‘Hepatobiliary disorders’ and ‘Investigations’, extracting their sub-hierarchical data from BioPortal. Under ‘Hepatobiliary disorders’, we selected five HLGT terms: ‘Hepatic and hepatobiliary disorders’, ‘Hepatobiliary neoplasm’, ‘Bile duct disorders’, ‘Gallbladder disorders’, and ‘Hepatobiliary investigations’. Additionally, to include laboratory blood tests for liver function, we chose the HLGT term ‘Hepatobiliary investigations’ under ‘Investigations’. We then utilized HLT and PT level terms within these selected HLGT terms to classify each reference in detail. Each HLT-PT set was paired to ensure precise clustering and annotation of data. To maintain focus on liver toxicity, we limited the inclusion of terms related to bile duct or gallbladder to one HLT-PT set per organ (*i.e.* ‘Bile duct disorders’- ‘Bile duct disorders’ and ‘Gallbladder disorders’- ‘Gallbladder disorders’). Furthermore, recognizing the clinical complexity, we selected the HLGT term ‘Hepatobiliary neoplasms’ to cover terms related to liver cancer at the PT level.

Calculation of the hepatotoxicity score

Due to the heterogeneous nature of information resources, estimating the reliability and relevance of records to hepatotoxicity poses challenges. To consolidate multiple records into a single metric, we developed a scoring system that assigns higher weights to clinical references over in vitro and in vitro data. In our classification of references, we assigned arbitrary weights of 3 for clinical evidence, 2 for in vivo evidence, and 1 for in vitro evidence. The overall hepatotoxicity

score for a compound (c) is calculated as the weighted sum of contributions from all records across nine source databases, taking into account whether each record has a positive or negative impact on hepatotoxicity:

$$S_c = \sum_{i=1}^{n_c} \text{sign}_c(i) \times \text{weight}_c(i) \quad (1)$$

where: n_c , number of records for compound c; $\text{sign}_c(i)$, +1 or−1 according to whether the record describes positive or negative evidence of hepatotoxicity; $\text{weight}_c(i)$, 3, 2, or 1 for clinical, in vivo, or in vitro evidence, respectively.

Conflicting records within a database are excluded from the sum (*i.e.*, given a weight of zero). This scoring system allows us to assess the overall hepatotoxicity potential of a compound based on aggregated evidence from diverse sources while considering the varying quality and type of data provided by each database.

HTP-KB contents and statistics

The integration of nine databases followed by manual curation and scoring has resulted in the creation of the most comprehensive knowledgebase on hepatotoxicity. We provide a brief overview of the statistics for the HTP-KB contents, including evidence classes, source databases, annotation levels, and overall hepatotoxicity scores in Fig. 2. Additionally, the detailed contributions and compound overlaps from each database are presented in Supplementary Fig. S3. All statistics are based on the PubChem CIDs.

HTP-KB includes a total of 8306 compounds curated manually into three classes by toxicology experts. There are 2260 (27.2%) entries supported by clinical evidence, significantly surpassing entries found in LiverTox or DILI-Irank (Fig. 2a and b). Entries supported by in vitro evidence constitute the largest portion, with 6472 (77.9%) compounds, indicating that HTP-KB has substantially broadened the scope of hepatotoxic compounds by incorporating in vitro evidence.

Analyzing the source databases of the records, 2260 entries in the clinical class are distributed across databases such as LiverTox (1005), T3DB (890), SIDER (748), and DILI (669) (Fig. 2b). CEBS contributes the largest collection of in vivo evidence, albeit representing a smaller portion of the knowledgebase. Almost all in vitro evidence is sourced from InvitroDB.

Next, we examine the distribution of the hepatotoxicity scores within our database, ranging from − 7 to +16 (Fig. 2c). The histogram plot showed a skewed distribution towards the positive side, likely because it is generally easier to determine positive hepatotoxicity compared to negative hepatotoxicity based on

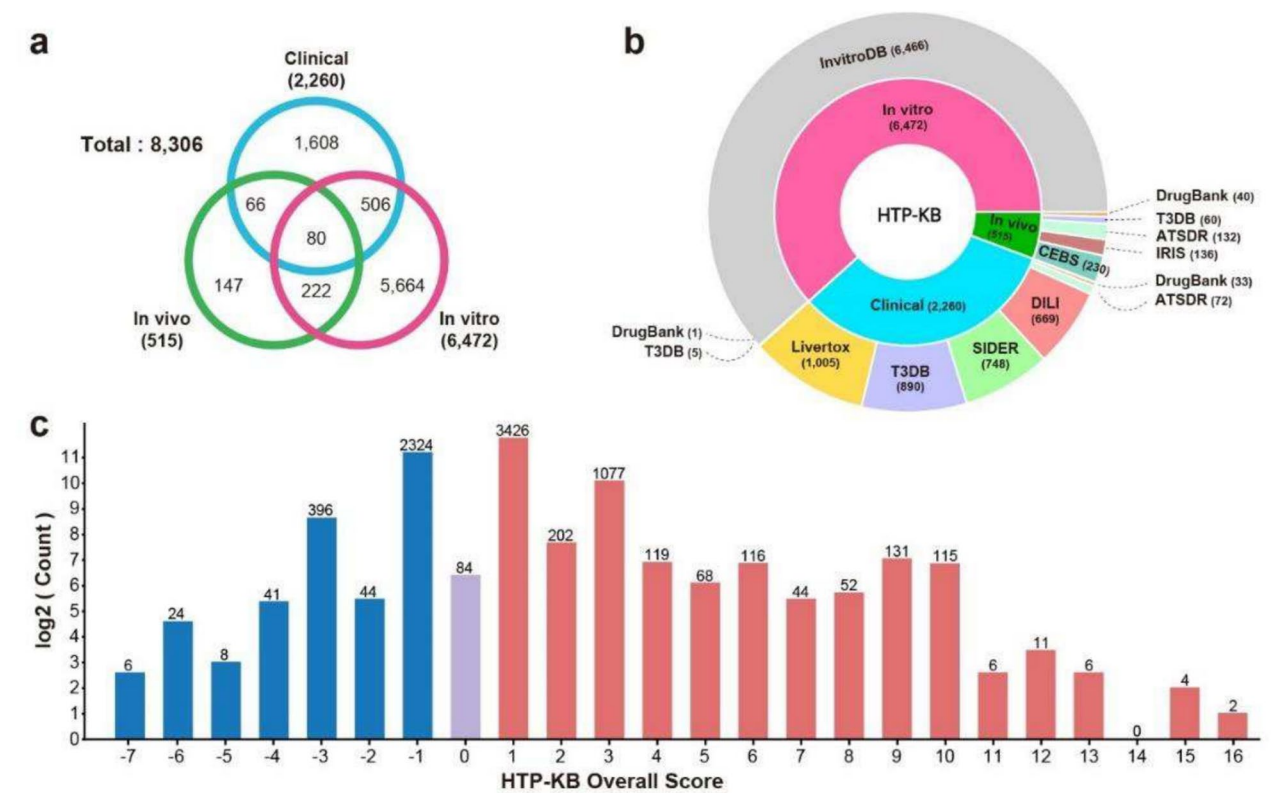


图2 HTP-KB数据统计。a 含有临床、体内和体外证据的化合物维恩图。b 不同类别的证据记录在各来源数据库中的分布。注意，每个化合物可能在多个数据库中被注释。c HTP-KB中所有化合物的整体肝毒性评分直方图。注意，频率值采用log2比例，以有效地可视化分布。

实验或文献证据。总体而言，HTP-KB 包含 5379 种在整体肝毒性方面得分为正的化合物，以及 2843 种得分为负的化合物。

使用 MedDRA 术语进行注释可以提供有关生物学功能的宝贵见解。我们的注释 o f 肝毒性利用了MedDRA术语中的高层次术语（HLT）和优先术语（PT）的组合。HLT术语中最大的一部分归因于“肝细胞损伤和肝炎 NEC”（30%），涵盖了各种PT术语，如“肝毒性”、“肝炎”、“肝损伤”和“肝坏死”用于子级分类（补充图S4）。其他重要的HLT术语包括“胆汁淤积和黄疸”（14%）、“肝酶和功能异常”（14%）以及“肝及肝胆疾病 NEC”（13%）。与癌症相关的术语如“腺瘤”和“癌”所占比例相对较小（分别为5%和5%）。

HTP-Pred模型的开发 HTP-KB数

据集的预处理 为了为HTP-Pred模型准备训练和测试数据，我们通过额外的预处理步骤进一步整理了原始的HTP-KB数据集。具体而言，在排除了重原子少于3个或多于60个的分子后，数据被重新标注为二分类，即肝毒性或非肝毒性化合物。合并不同的肝毒性数据集常常会导致数据条目标签冲突。排除所有此类条目会因训练数据不足或有限数据过拟合而对模型性能产生不利影响。为了解决这一问题，我们通过优先考虑数据源数据库的可靠性顺序（按临床、体内、体外的顺序）来解决标签冲突。此外，当化合物的证据在相同可靠性水平上相互矛盾时，我们排除了模糊案例。这种方法确保模型在训练时使用高置信度的数据，同时保持足够的数据量。对于

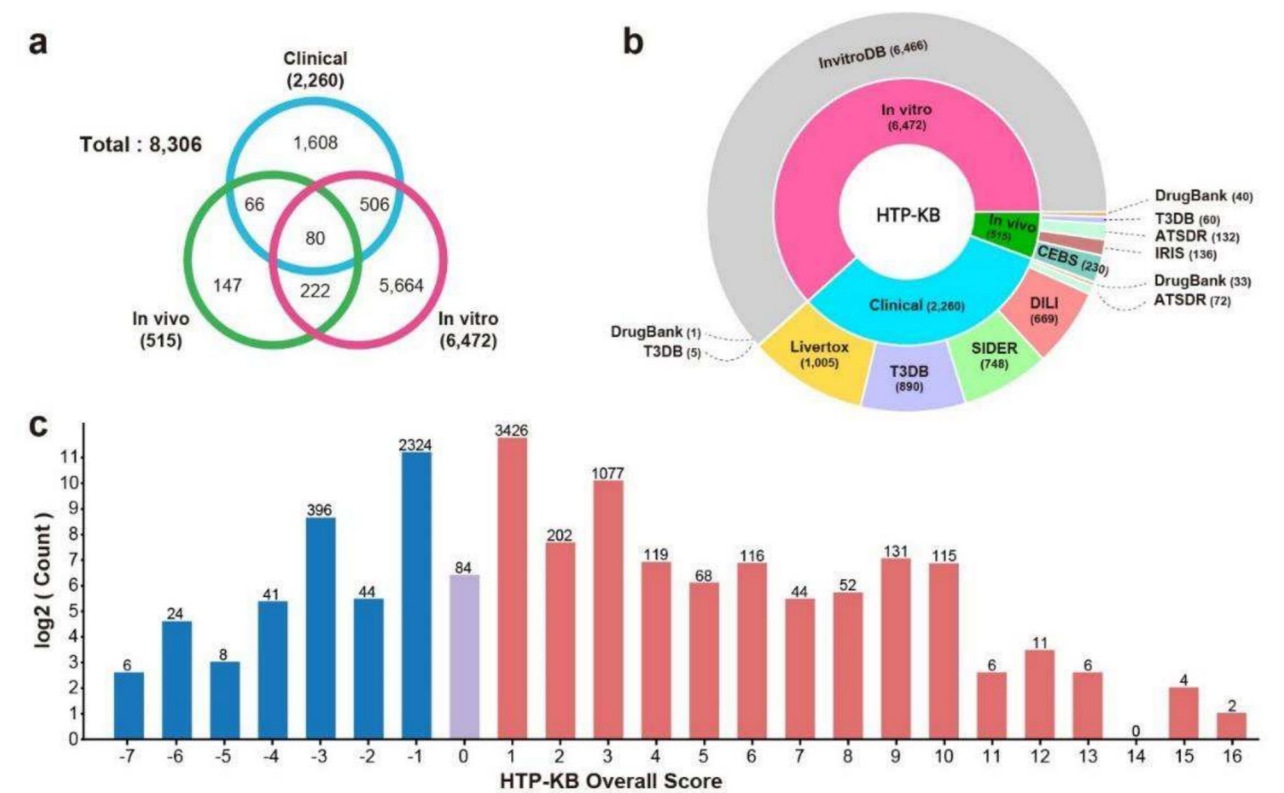


Fig. 2 Statistics of HTP-KB data. **a** Venn diagram of compounds with clinical, in vivo, and in vitro evidences. **b** Distribution of different classes of evidence records across source databases. Note that each compound may be annotated in multiple databases. **c** Histogram of overall hepatotoxicity score for all compounds in HTP-KB. Note that the frequency values are on a log2 scale to visualize the distribution effectively

experimental or literature evidence. Overall, HTP-KB includes 5379 compounds with positive scores and 2843 compounds with negative scores in terms of overall hepatotoxicity.

Annotation using MedDRA terms provides valuable insights into biological functions. Our annotation of hepatotoxicity utilizes a combination of High-Level Term (HLT) and Preferred Term (PT) terms from MedDRA terminology. The largest portion of the HLT terms is attributed to ‘Hepatocellular damage and hepatitis NEC’ (30%), encompassing various PT terms such as ‘Hepatotoxicity’, ‘Hepatitis’, ‘Liver injury’, and ‘Hepatic necrosis’ for sub-level categorizations (Supplementary Fig. S4). Other significant HLT terms include ‘Cholestasis and jaundice’ (14%), ‘Hepatic enzymes and function abnormalities’ (14%), and ‘Hepatic and hepatobiliary disorders NEC’ (13%). Cancer-related terms such as ‘adenoma’ and ‘carcinoma’ contributed to a relatively small portion (5% and 5%, respectively).

Development of HTP-Pred model

Pre-processing the HTP-KB dataset

To prepare the training and test data for the HTP-Pred model, we further curated the original HTP-KB dataset through additional pre-processing steps. Specifically, the data were re-labeled into binary classes as either hepatotoxic or non-hepatotoxic compounds after excluding molecules with fewer than three or more than 60 heavy atoms. Merging diverse hepatotoxicity datasets often results in data entries with conflicting labels. Excluding all such entries affects the model performance adversely due to insufficient amount of training data or overfitting limited amount of data. To address this, we resolved label conflicts by prioritizing the source database in the following order of reliability: clinical, in vivo, and in vitro. Additionally, we excluded ambiguous cases when the evidence for a compound contradicts each other at the same level of reliability. This approach ensures the model is trained on higher-confidence data while maintaining a sufficient number of data points. For evaluating robustness of the model upon imbalanced dataset, we employed stratified tenfold cross-validation to calculate the average performance score and standard deviation. Additionally, for

为了与其他肝毒性预测工具进行比较, 我们进行了留出验证。数据集按8:1:1的比例划分为训练集、验证集和测试集, 同时保持阳性与阴性类别分布的一致性。此次划分导致训练集中有5592个化合物, 验证集中有699个, 测试集中有700个。

使用 *HTP-KB* 数据集微调 *MolCLR*

接下来, 我们通过微调一个预训练的图神经网络 (GNN) 模型开发了一个肝毒性分类模型 (图3)。在大量数据上预训练的深度学习模型被广泛用作各种下游任务的基础框架, 尤其是在标注数据有限的情况下 [44, 45]。肝毒性预测就是这样一种情况; 尽管从不同数据库进行了严格的数据整理, 仅在 *HTP-KB* 数据集上训练模型还不足以覆盖广泛的化学空间。为了解决这一限制, 我们采用了 *MolCLR* [46], 这是一个利用自监督学习技术预训练的 GNN。*MolCLR* 利用来自 PubChem 的大约 1000 万个独特分子进行对比学习任务, 从而能够学习具有通用性的分子特征。

表示。这种方法使模型能够适应分子性质预测的下游任务, 在回归和分类基准测试中均表现出优越的性能。因此, 我们在 *HTP-KB* 数据集上对 *MolCLR* 的基础 GNN 模型进行了微调, 以弥补数据稀缺并增强对肝毒性的预测能力。

我们使用图卷积网络 (GCN) [47] 或图同构网络 (GIN) [48] 作为预训练模型的 GNN 骨干, 预训练参数由原始 *MolCLR* 实现提供。对于二分类任务, 我们在预训练的 GNN 特征提取模块上附加了一个随机初始化的多层感知器 (MLP) 预测头。根据 *MolCLR* 的训练协议, 我们对模型进行了 100 轮的微调, 基模型的初始学习率为 1×10^{-4} , 预测头的初始学习率为 5×10^{-4} 。微调后的模型命名为 *HTP-Pred*。

HTP-Pred 的性能总结如表 2 所示。作为基线, 我们使用来自 InterDILI [49] 的分子描述符来构建输入特征, 并应用包括支持向量机在内的机器学习 (ML) 方法

comparison with other hepatotoxicity prediction tools, we conducted hold-out validation. The dataset was split into training, validation, and test sets in an 8:1:1 ratio, maintaining an equivalent positive-to-negative class distribution. This split resulted in 5592 compounds in the training set, 699 in the validation set, and 700 in the test set.

Fine-tuning *MolCLR* with the *HTP-KB* dataset

Next, we developed a hepatotoxicity classification model by fine-tuning a pre-trained graph neural network (GNN) model (Fig. 3). Pre-trained deep learning models on large amount of data are widely employed as foundational frameworks for various downstream tasks, particularly in cases with limited labeled data [44, 45]. Hepatotoxicity prediction is one such case; despite rigorous data curation from diverse databases, training a model solely on the *HTP-KB* dataset is insufficient to capture a broad chemical space. To address this limitation, we employed *MolCLR* [46], a pre-trained GNN utilizing self-supervised learning techniques. *MolCLR* leverages approximately 10 million unique molecules from PubChem for contrastive learning task, enabling it to learn generalizable molecular

representations. This approach allows the model to adapt to downstream tasks of molecular property prediction, demonstrating superior performance on both regression and classification benchmarks. Accordingly, we fine-tuned the base GNN model of *MolCLR* on the *HTP-KB* dataset, compensating for data scarcity and enhancing hepatotoxicity prediction.

We utilized either a graph convolutional network (GCN) [47] or graph isomorphism network (GIN) [48] as the GNN backbone for the pre-trained model, with pre-trained parameters provided by the original *MolCLR* implementation. For the binary classification task, we appended a randomly initialized multi-layer perceptron (MLP) prediction head to the pre-trained GNN feature extractor module. Following *MolCLR*'s training protocol, we fine-tuned the model for 100 epochs, using an initial learning rate of 1×10^{-4} for the base model and 5×10^{-4} for the prediction head. The resulting fine-tuned model was named *HTP-Pred*.

The performance of *HTP-Pred* is summarized in Table 2. As baselines, we used molecular descriptors from InterDILI [49] to build input features and applied machine learning (ML) methods, including support

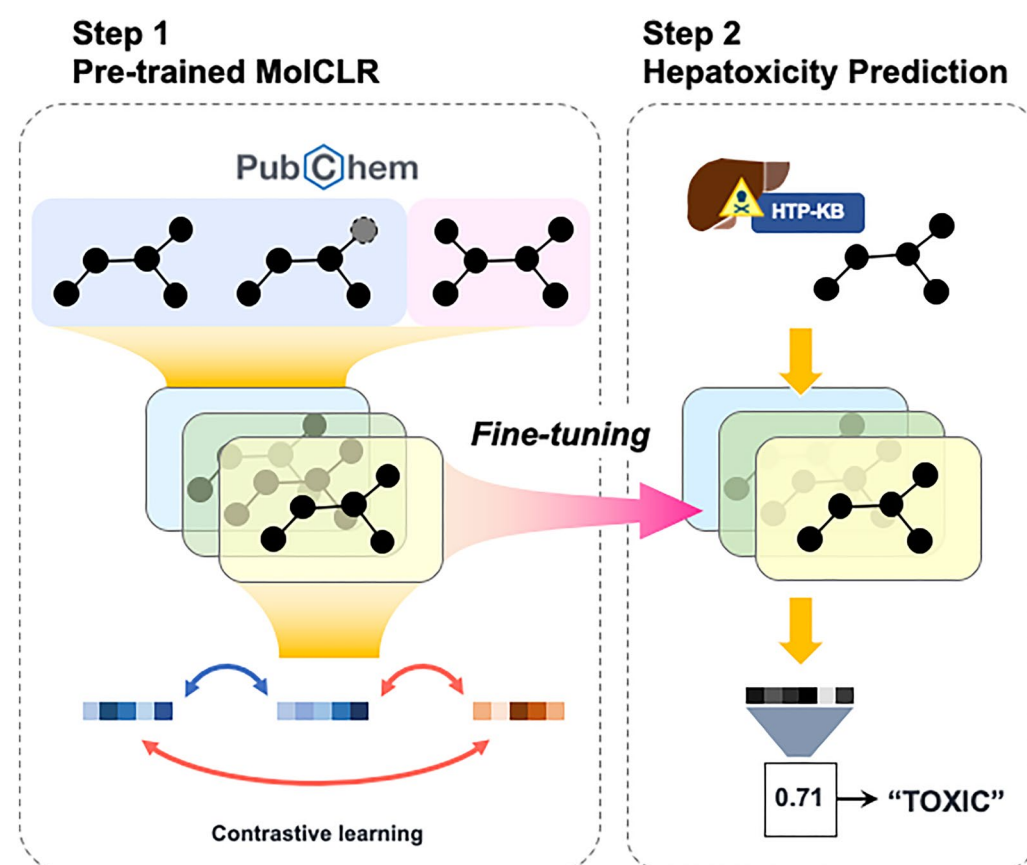


图 3 *HTP-Pred* 模型结构

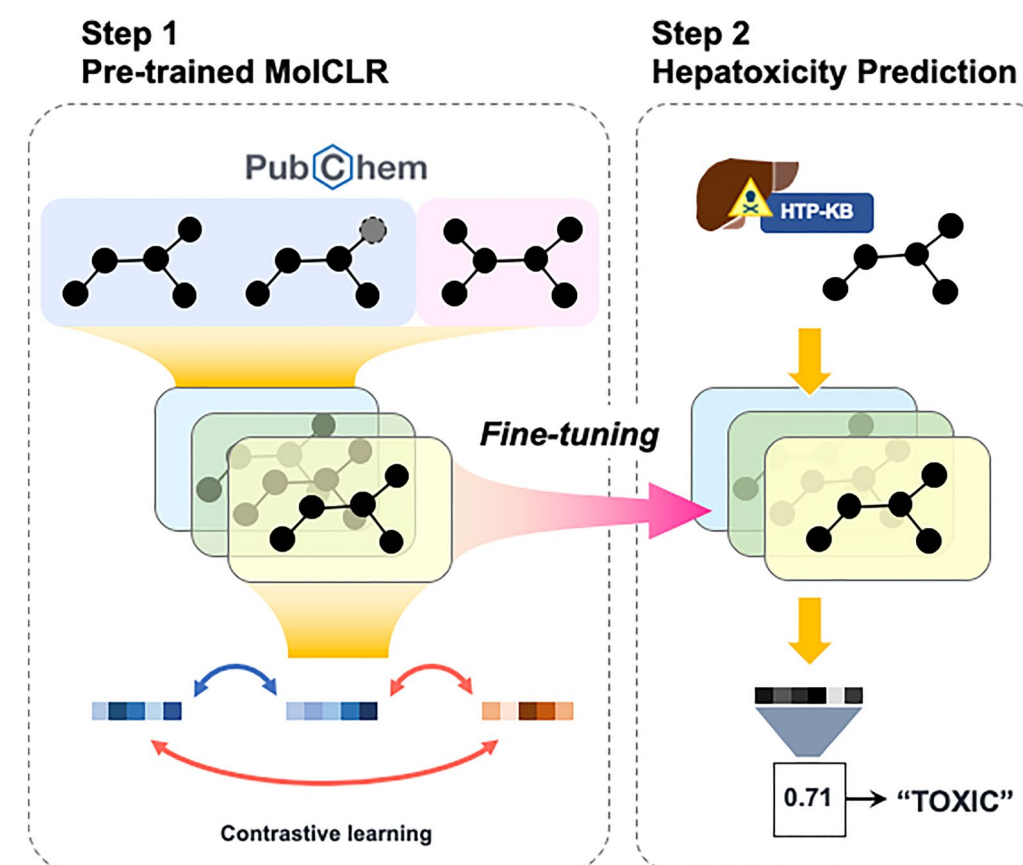


Fig. 3 Structure of *HTP-Pred* model

表2 基于机器学习的基线模型和采用不同预训练方法的 HTP-Pred模型的肝毒性预测性能, 使用分层十折交叉验证

模型	预训练方法	受试者工作特征曲线下面积
SVM (线性)	None	0.638 ± 0.015
支持向量机 (多项式)	None	0.538 ± 0.009
支持向量机 (RBF)	None	0.658 ± 0.017
RF	None	0.677 ± 0.011
逻辑回归	None	0.633 ± 0.022
HTP –Pred – GCN	None	0.755 ± 0.020
	MolCLR	0.740 ± 0.023
HTP - Pred - GIN	None	0.763 ± 0.021
	MolCLR	0.772 ± 0.019

最佳AUROC得分以加粗显示

向量机 (SVM)、随机森林 (RF) 和逻辑回归用于分类。对于SVM, 我们测试了三种核类型: 线性、多项式和径向基函数 (RBF)。此外, 我们通过从头训练骨干模型进行了预训练消融研究。我们还比较了基于GCN和GIN的预训练模型的性能。使用AUROC分数作为评估指标, 它能在不同阈值下捕捉二分类性能。在基于机器学习的方法中, RF取得了最佳性能, 这与InterDILI的结果一致。然而, 即使没有预训练, 基于GNN的分类器在AUROC分数上也优于基线的ML模型。在两个骨干网络中, GIN始终优于GCN。对MolCLR进行微调进一步提高了基于GIN的性能, 达到了最佳AUROC分数 0.772。这些结果表明, 基于GIN的预训练MolCLR有效

捕获信息丰富的分子表示, 从而实现优越的肝毒性预测。

接下来, 我们使用在留出验证集上训练的模型, 评估了HTP-Pred预测结果与HTP-KB肝毒性整理评分之间的一致性。根据肝毒性评分, HTP-KB中的化合物被分为三组: 肝毒性阴性 (KB评分: − 7 到 0)、中度阳性 (KB评分: 0到7) 和高阳性 (KB评分: >7)。阴性组中的化合物其HTP-Pred评分显著低于阳性组化合物, 表明HTP-Pred能够有效区分肝毒性阴性化合物与肝毒性阳性化合物 (补充图S5)。然而, 中度阳性组与高阳性组的评分分布相似, 这可能是因为模型训练的目标是预测肝毒性的有无二分类, 而非具体评分值。

此外, 我们将 HTP-Pred 的性能与以往用于化合物肝毒性预测的工具进行了比较 (表 3)。尽管我们旨在使用完整的 700 个化合物测试集, 但由于一些工具在输入方面存在限制, 比较范围仅限于 644 个重叠化合物。这些化合物的列表可在模型仓库中获得, 同时提供了模型脚本 (https://github.com/WonhoZhung/HTP_Pred)。结果表明, HTP-Pred 的表现优于现有的毒性预测工具, 这很可能归因于经过严格整理的数据集与先进的深度学习技术的结合, 包括基于 GIN 的分子表征以及对在大型无标签数据集上预训练的 GNN 模型进行微调。有关此比较分析的更多细节, 可以在补充材料中找到。

表3 使用644种重叠化合物与现有预测工具的性能比较

模型 (预测工具)	肝脏毒性数据	算法	准确性	受试者工作特征曲线下面积	参考文献
ProTox [37] 3 (网络服务器)	肝毒	RF	0.457	0.549	[37]
毒性星 (网页服务器)	DILI排名, 药典数据库	SVM, RF	0.534	0.495	[31]
eMolTox (网页服务器)	38/174 期刊调查数据集	RF	0.598	0.553	[39]
admetSAR (网页服务器)	— ^a	— ^a	0.624	0.498	[38]
肝舒乐-EL (网页服务器)	DILIrank, FDA橙皮书	合奏 (随机森林, 支持向量机, 极端梯度提升)	0.618	0.558	[29]
深度药物性肝损伤 (基于代码)	DILlst, DrugBank、维基百科、FDALabelDB	合奏 (LR, RNN, XGB, RF, SVM)	0.620	0.586	[32]
HTP-预测	9 个数据集	GIN	0.717	0.761	

^a 无法指定肝脏特异训练数据集或模型算法

Table 2 Hepatotoxicity prediction performance of ML-based baseline models and HTP-Pred models with different pre-training method, with stratified tenfold cross-validation

Models	Pre-training method	AUROC
SVM (linear)	None	0.638 ± 0.015
SVM (polynomial)	None	0.538 ± 0.009
SVM (RBF)	None	0.658 ± 0.017
RF	None	0.677 ± 0.011
Logistic regression	None	0.633 ± 0.022
HTP-Pred-GCN	None	0.755 ± 0.020
	MolCLR	0.740 ± 0.023
HTP-Pred-GIN	None	0.763 ± 0.021
	MolCLR	0.772 ± 0.019

Best AUROC score shown in bold

vector machine (SVM), random forest (RF), and logistic regression, for classification. For SVM, we tested three kernel types: linear, polynomial, and radial basis function (RBF). Additionally, we conducted an ablation study on pre-training by training the backbone model from scratch. We also compared the performance of GCN- and GIN-based pre-trained models. AUROC scores were used as an evaluation metric, which captures the binary classification performance across different thresholds. Among the ML-based methods, RF achieved the best performance, consistent with the results from InterDILI. However, even without pre-training, the GNN-based classifiers outperformed the baseline ML models in terms of AUROC scores. Between the two backbones, GIN consistently outperformed GCN. Fine-tuning MolCLR further improved GIN-based performance, achieving the best AUROC score of 0.772. These results demonstrate that the pre-trained GIN-based MolCLR effectively

captures informative molecular representations, leading to superior hepatotoxicity prediction.

Next, we evaluated the concordance between HTP-Pred predictions and the hepatotoxicity curation scores from HTP-KB, using the model trained on the hold-out validation set. Compounds in HTP-KB were categorized into three groups based on their hepatotoxicity scores: hepatotoxicity negative (KB score: − 7 to 0), moderately positive (KB score: 0 to 7), and highly positive (KB score:>7). Compounds in the negative group exhibited significantly lower HTP-Pred scores compared to those in the positive groups, indicating that HTP-Pred effectively distinguishes hepatotoxicity-negative compounds from hepatotoxicity-positive ones (Supplementary Fig. S5). However, the moderately positive and highly positive groups showed similar score distributions, likely because the model was trained to predict the binary presence or absence of hepatotoxicity rather than specific score values.

Additionally, we compared HTP-Pred’s performance against previous liver toxicity prediction tools for compounds (Table 3). Although we aimed to use the full test set of 700 compounds, some tools were limited by input constraints, restricting the comparison to 644 overlapping compounds. The list of these compounds is available in the model repository, alongside the model scripts (https://github.com/WonhoZhung/HTP_Pred). The results demonstrate that HTP-Pred outperforms existing toxicity prediction tools, likely due to the combination of a robustly curated dataset and advanced deep learning techniques, including the GIN-based molecular representation and fine-tuning of a GNN model pre-trained on large, unlabeled datasets. Further details on this comparative analysis can be found in the Supplementary materials.

Table 3 Performance comparison with existing prediction tools using 644 overlapping compounds

Models (Prediction tool)	Liver toxicity data	Algorithm	Accuracy	AUROC	References
ProTox [37] 3 (Webserver)	LiverTox	RF	0.457	0.549	[37]
ToxSTAR (Webserver)	DILIrank, PharmaPendium	SVM, RF	0.534	0.495	[31]
eMolTox (Webserver)	38/174 journal survey dataset	RF	0.598	0.553	[39]
admetSAR (Webserver)	— ^a	— ^a	0.624	0.498	[38]
Hepatopred-EL (Webserver)	DILIrank, FDA orange book	Ensemble (RF, SVM, XGB)	0.618	0.558	[29]
DeepDILI (Code-based)	DILlst, DrugBank, Wikipedia, FDALabelDB	Ensemble (LR, KNN, XGB, RF, SVM)	0.620	0.586	[32]
HTP-Pred	9 Datasets	GIN	0.717	0.761	

^a Unable to specify the liver-specific training dataset or model algorithm

肝毒性基团计算

为了增强我们对小分子肝毒性预测的理解和可解释性, 有必要识别分子中各个原子或子结构的贡献。基于梯度的方法 [50, 51], 最初用于评估基于图像的预测中像素的贡献, 已被改编用于 HTP-Pred 模型。对于给定的分子图 $\mathcal{G}$, 每个原子 a 被表示为节点特征 $X_a \in \mathbb{R}^F$, 其中 F 表示特征维度。为了确定每个原子对预测的贡献, 我们首先计算预测输出 $y_{\mathcal{G}}$ 对输入节点特征的绝对梯度:

$$\tilde{c}(\mathcal{G}, a) = \sum_{i=1}^F \left| \frac{\partial y_{\mathcal{G}}}{\partial X_{a,i}} \right|,$$

其中 $\tilde{c}(\mathcal{G}, a)$ 表示原子 a 的未归一化贡献分数。接下来, 这些分数在分子中的所有原子之间进行归一化, 以获得原子贡献分数 $c(\mathcal{G}, a)$:

$$c(\mathcal{G}, a) = \frac{\tilde{c}(\mathcal{G}, a)}{\sum_b \tilde{c}(\mathcal{G}, b)}.$$

这种方法量化了每个原子或子结构对模型预测结果的贡献, 从而能够识别输入分子中的肝毒性基团 (有毒子结构)。请注意, 未归一化的贡献得分为正, 因此归一化的原子贡献得分介于0和1之间。

为了定义毒性基团, 我们使用了一组来自Yang等人的SMiles任意靶标规范 (SMARTS) 模式 [52], 这些模式使用化学信息学语言来描述化学模式。使用 RDKit函数在每个化合物中搜索这些子结构模式。之前获得的原子贡献分数被汇总到每个模式对应的原子, 以推导出每个子结构的总体分数 $c(\mathcal{G}, S)$ S 。

$$c(\mathcal{G}, S) = \sum_{a \in V(S)} c(\mathcal{G}, a),$$

其中 $V(S)$ 表示包含子结构 S 的原子集合。通过毒性原基 SMARTS 匹配识别的子结构得分也可以从 0 到 1 变化, 表示该子结构对模型决策的贡献。该方法通过根据总体得分对子结构进行排名, 从而实现了关键毒性原基的识别。这些排名后的毒性原基为肝毒性预测中最关键的分子特征提供了见解。

HTP 数据库和网页服务器实现

数据库建设

PubChem CID 被用作每种化合物的主要标识符, 以便高效地将各个数据库的具体内容与整体整理汇总结果连接起来。此外, 为各自数据库的参考资料创建了样本 ID, 格式为以数据库名称缩写为前缀的数字标识符。编译了一个 SQL 文件, 将所有数据库样本 ID 与主 PubChem CID 整合在一起, 同时纳入附加的分子属性和相应的 HTP-Pred 结果。该网站服务器通过查询这个综合的 SQL 文件来运行, 从而确保对综合数据的无缝访问。

网页界面概览

HTP 网络界面旨在为用户提供有关化学肝毒性的信息, 使其易于访问且内容全面。“搜索”部分允许用户通过多种方式识别化合物, 支持多种化学物质识别格式, 并提供化学结构的可视化呈现, 以增强可用性。集成的统计页面呈现了数据集的摘要, 为用户提供肝毒性数据覆盖的全面和详细视图。“下载”部分允许用户下载整个整理的数据集或单个数据库的特定子集, 从而进行进一步分析和研究。为了帮助用户有效地导航和使用 HTP 网络服务器, “帮助”页面提供了详细的说明和使用指南。这个用户友好的界面确保研究和探索肝毒性数据时能够简化访问流程。

复合搜索和浏览

在“搜索”模块中, 用户可以通过查询化合物ID或绘制化学结构来搜索化合物 (图4)。除了PubChem CID 这一主要标识符外, 我们还支持多种ID格式, 例如通用化合物名称、IUPAC名称、SMILES、CASID和分子式。用户可以选择精确匹配的结果, 或根据需要探索结构相似或子结构化合物。此外, 用户可以使用JSME分子编辑器输入自己的原始分子。在HTP-KB中未找到匹配化合物的情况下, 仅显示HTP-Pred结果, 该结果将在结果界面部分进一步详细说明。

另外, 用户可以利用“统计”模块来探索各个数据库的总体数据并选择偏好的化合物。虽然此页面提供了我们数据的全面统计信息, 但点击每个数据库名称会引导用户进入详细的数据浏览表。结果表包含用户友好的筛选选项。

Hepato-toxicophore calculation

To enhance our comprehension and explainability of hepatotoxicity predictions for small molecules, it is essential to identify the contributions of individual atoms or substructures within a molecule. Gradient-based methods [50, 51], originally developed to assess pixel contributions in image-based predictions, were adapted for use with the HTP-Pred model. For a given molecular graph $\mathcal{G}$, each atom a is represented as a node feature $X_a \in \mathbb{R}^F$, where F denotes the feature dimension. To determine the contribution of each atom to the prediction, we first compute the absolute gradient of the prediction output $y_{\mathcal{G}}$, with respect to the input node features:

$$\tilde{c}(\mathcal{G}, a) = \sum_{i=1}^F \left| \frac{\partial y_{\mathcal{G}}}{\partial X_{a,i}} \right|,$$

where $\tilde{c}(\mathcal{G}, a)$ represents the unnormalized contribution score for atom a . Next, these scores are normalized across all atoms in the molecule to obtain the atom contribution score $c(\mathcal{G}, a)$:

$$c(\mathcal{G}, a) = \frac{\tilde{c}(\mathcal{G}, a)}{\sum_b \tilde{c}(\mathcal{G}, b)}.$$

This approach quantifies the contribution of each atom or substructure to the model's prediction outcome, enabling the identification of hepato-toxicophores (toxic substructures) within the input molecule. Note that the unnormalized contribution score is positive, so the normalized atom contribution score ranges between 0 and 1.

To define toxicophores, we used a set of SMiles Arbitrary Target Specification (SMARTS) patterns derived from Yang et al. [52], which employ a cheminformatics language for describing chemical patterns. RDKit functions were utilized to search for these substructure patterns within each compound. Atom contribution scores obtained earlier were summed for each pattern's corresponding atoms to derive an overall score $c(\mathcal{G}, S)$ for each substructure S :

$$c(\mathcal{G}, S) = \sum_{a \in V(S)} c(\mathcal{G}, a),$$

where $V(S)$ represents the set of atoms comprising substructure S . The score of substructures, identified through toxicophore SMARTS matching, can also range from 0 to 1, indicating the contribution of the substructure to the model's decision. This methodology enabled the identification of key toxicophores by ranking substructures based on their overall scores. These ranked toxicophores provide insights into the molecular features most critical for hepatotoxicity prediction.

HTP Database and web server implementation

Database construction

PubChem CID was utilized as the primary identifier for each compound to efficiently link specific contents from individual databases with overall curation summary results. Additionally, sample IDs were created for references from their respective databases, formatted as numeric identifiers prefixed with the abbreviated database name. An SQL file was compiled to consolidate all database sample IDs with the main PubChem CID, integrating additional molecular properties and the corresponding HTP-Pred results. The web server operates by querying this comprehensive SQL file, ensuring seamless access to integrated data.

Web interface overview

The HTP web interface is designed to provide users with accessible and comprehensive information on chemical hepatotoxicity. The ‘Search’ section allows users to identify compounds through various methods, supporting multiple chemical ID formats and featuring visual representations of chemical structures for enhanced usability. An integrated statistics page presents a summary of the dataset, offering users a broad and detailed view of hepatotoxicity data coverage. The ‘Downloads’ section allows users to download the entire curated dataset or specific subsets from individual databases, enabling further analysis and research. To assist users in navigating and utilizing the HTP web server effectively, detailed instructions and usage guidelines are provided on the ‘Help’ page. This user-friendly interface ensures streamlined access to hepatotoxicity data for research and exploration.

Compound searching and browsing

In the ‘Search’ module, users can search for chemical compounds either by querying compound IDs or by drawing chemical structures (Fig. 4). Alongside PubChem CID, the primary identifier, we support diverse ID formats such as general compound names, IUPAC names, SMILES, CASIDs, and molecular formulas. Users have the option to choose between exact matching results or explore structurally similar or substructural compounds as per their needs. Additionally, users can input their original molecules using the JSME molecule editor. In cases where no matching compound is found in HTP-KB, only the HTP-Pred result is displayed, which is further detailed in the result interface section.

Alternatively, users can utilize the ‘Statistics’ module to explore overall data across each database and select preferred compounds. While this page provides comprehensive statistics for our data, clicking on each database name directs users to a detailed data browsing table. The result table includes user-friendly filtering options

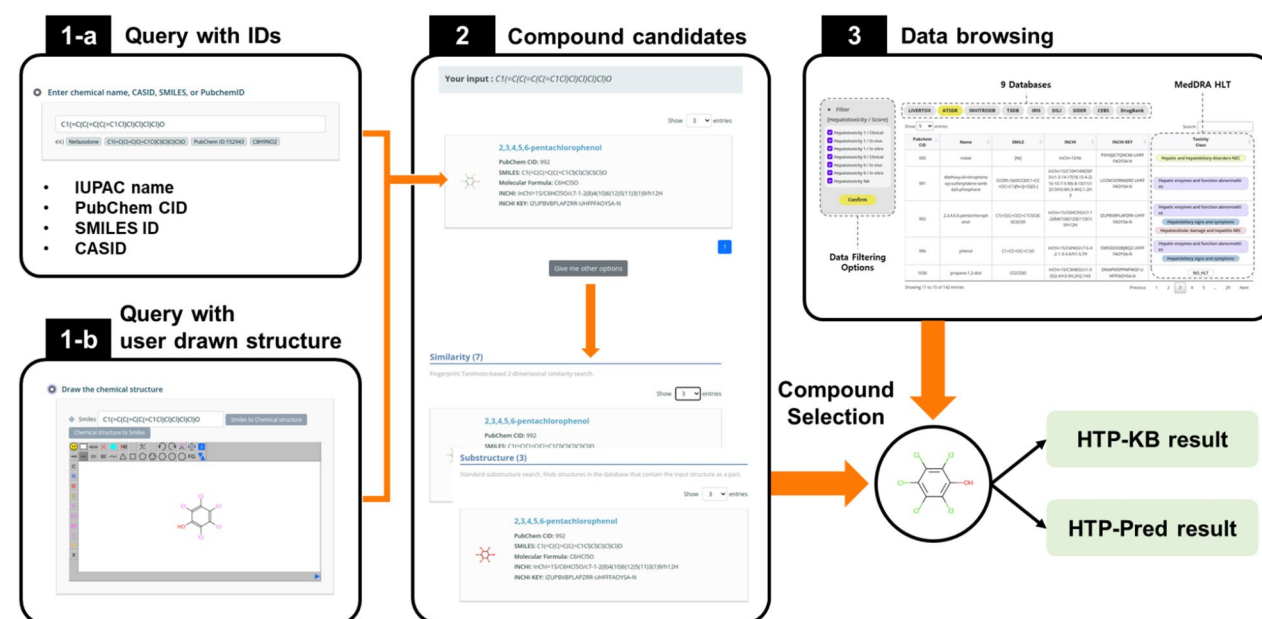


图4 化合物搜索的用户界面。该图展示了搜索和浏览化合物的示例页面。用户可以使用多种ID类型或JSME分子编辑器查询化合物。搜索结果界面允许用户通过各种选项选择完全匹配的化合物、相似化合物和子结构化合物。“统计”菜单提供访问各个数据库浏览表的功能，允许用户按毒性分类和评分选项进行筛选。它还包括每种化合物的基本ID信息和分子属性。在最终选择后，用户将看到两个主要页面：HTP-KB搜索结果和HTP-Pred结果。

通过位于表格旁边的选择栏，允许用户获取每个数据库中化合物的筛选列表。每个表格条目包含基本标识符，如 PubChem CID、SMILES、InChI 和 InChI Key，以及所有匹配的高级术语（HLT）的唯一列表。点击任意行会导航用户到特定的化合物结果页面。

HTP-KB 结果页面

所查询化合物的HTP-KB结果由几个活跃的子页面组成(图5)。在每个HTP-KB子页面的顶部,彩色条表示所查询化合物相对于分数分布的总体肝毒性评分。

在页面中央，一个主表格允许用户快速评估来自每个数据库的肝毒性参考文献，以及它们对应的重要性类别。表格中的彩色单元格表示数据的特性：红色表示肝毒性，蓝色表示非肝毒性。点击每个激活的单元格将在屏幕底部显示详细结果。

由于用于肝毒性判断的特征和证据信息不同，每个数据库子页面的格式各不相同。然而，所有子页面都包括指向原始数据库网页服务器的链接和注释的 MedDRA 毒性分类术语。

顶部。即使在单个数据库中，也可能提供多个参考按钮以显示对应不同MedDRA术语的结果。点击这些按钮会显示用于MedDRA术语决策的主要证据句子和总体数据。对于提供PDF格式文件作为资源的数据库，如ATSDR、DILI、LiverTox和IRIS，优先显示包含相关句子的页面，用户可以通过滚动嵌入的PDF文件访问其他页面。如果数据库的主要数据文件为CSV格式（例如CEBS和InvitroDB），子页面会呈现一个带有可选列的表格。初始情况下显示预先选择的列，但用户可以通过选择感兴趣的列自定义视图。一些数据库遵循上述未涵盖的不同格式。例如，T3DB在其子页面的多个部分中突出显示与数据决策相关的重要句子，而SIDER提供所有与MedDRA相关的参考文件。DrugBank仅呈现用于i的关键句子

HTP-Pred 结果页面

HTP的另一个重要输出是由HTP-Pred模块生成的预测结果（图6）。图中右上部分显示的主要毒性预测分数，表示肝毒性的可能性。该分数在显示分布的图上以绿色虚线表示

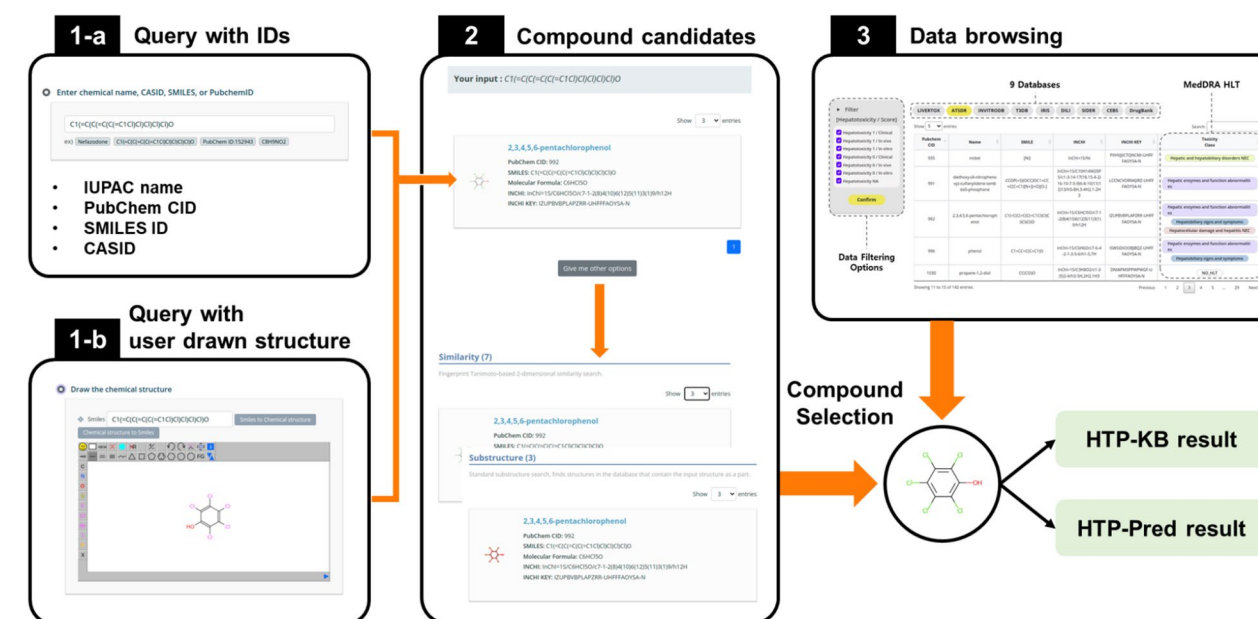


Fig. 4 User interface for compound searches. The figure illustrates example pages for searching and browsing compounds. Users can query compounds using several ID types or the JSME molecular editor. The search results screen allows users to select exact matching compounds, similar compounds, and substructural compounds through various options. The 'Statistics' menu provides access to individual database-wise browsing tables, allowing users to filter by toxicity class and score options. It also includes basic ID information and molecular properties for each compound. Upon final selection, users are presented with two main pages: HTP-KB search results and HTP-Pred results

via a selection bar adjacent to the table, allowing users to obtain a filtered list of compounds within each database. Each table entry features basic identifiers such as PubChem CID, SMILES, InChI, and InChI Key, alongside all unique lists of matched High-Level Terms (HLT). Clicking on any row navigates users to the specific compound result page.

HTP-KB result page

The HTP-KB result for the queried compound consists of several active subpages (Fig. 5). At the top of each HTP-KB subpage, a color bar indicates the overall hepatotoxicity score of the queried compound relative to the score distribution.

In the center of the page, a main table allows users to quickly assess the hepatotoxicity references from each database, along with their corresponding importance classes. Colored compartments within the table signify the characteristics of the data: red for hepatotoxic and blue for non-hepatotoxic. Clicking on each activated compartment reveals detailed results at the bottom of the screen.

Each database subpage varies in format due to distinct characteristics and evidence information for hepatotoxicity determination. However, all subpages include links to the original database web server and annotated MedDRA toxicity classification terms at the

top. Even within a single database, multiple reference buttons may be provided to display results corresponding to various MedDRA terms. Clicking these buttons shows the main evidence sentence and overall data used for MedDRA term decisions. For databases such as ATSDR, DILI, LiverTox, and IRIS, which offer PDF-formatted files as resources, pages containing relevant sentences are prioritized, with additional pages accessible by scrolling through the embedded PDF file. If a database's primary data file is in CSV format (e.g., CEBs and InvitroDB), a subpage presents a table with selectable columns. Initially, pre-selected columns are displayed, but users can customize the view by selecting columns of interest. Some databases follow different formats not covered above. For instance, T3DB highlights crucial sentences related to data decisions among multiple sections on its subpages, while SIDER provides all MedDRA-related reference files. DrugBank presents only the critical sentence used in toxicity determination directly.

HTP-Pred result page

Another significant output of HTP is the prediction result generated by the HTP-Pred module (Fig. 6). The primary toxicity prediction score, displayed at the upper right part of the figure, indicates the likelihood of hepatotoxicity. This score is represented as a green dotted line on a plot showing the distribution of

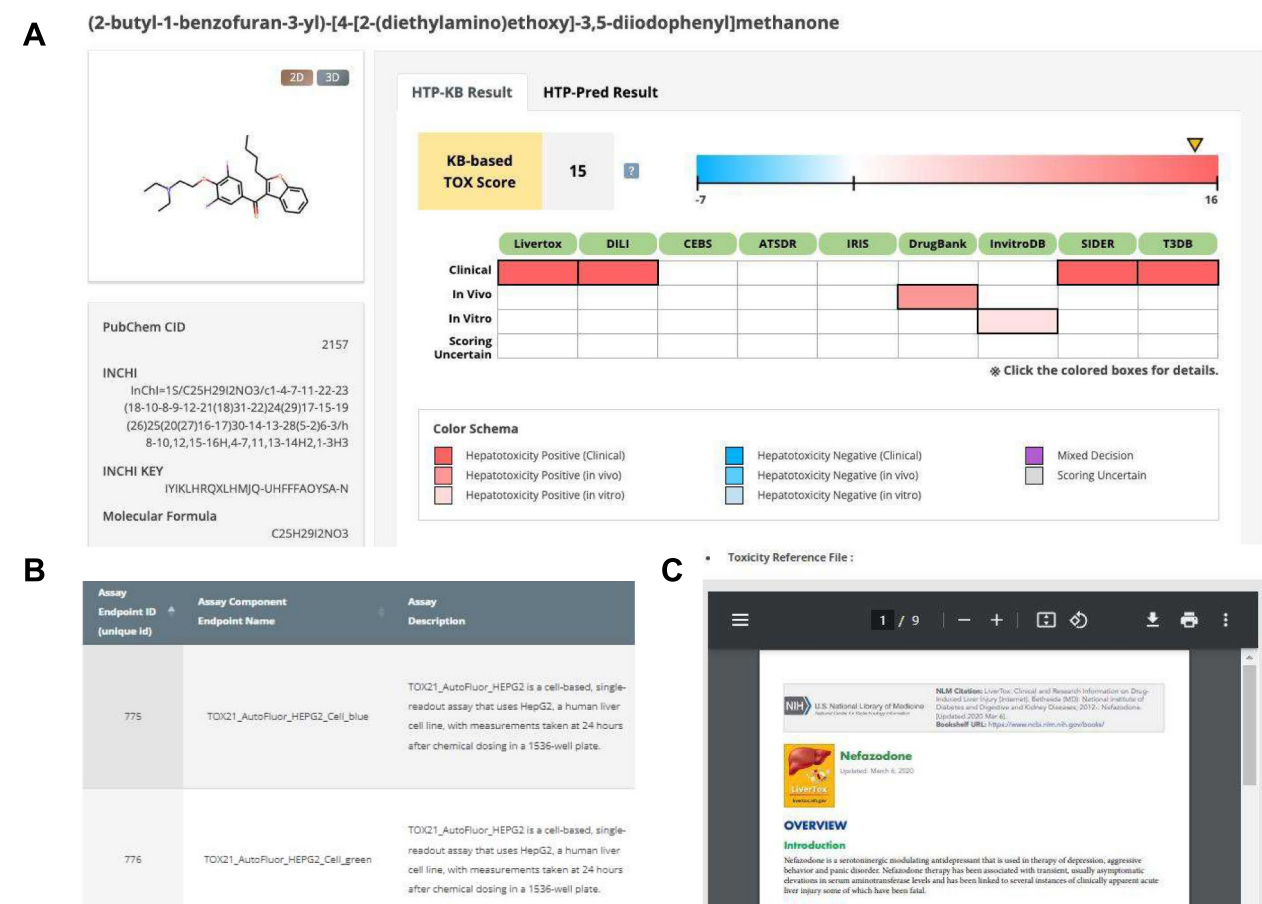
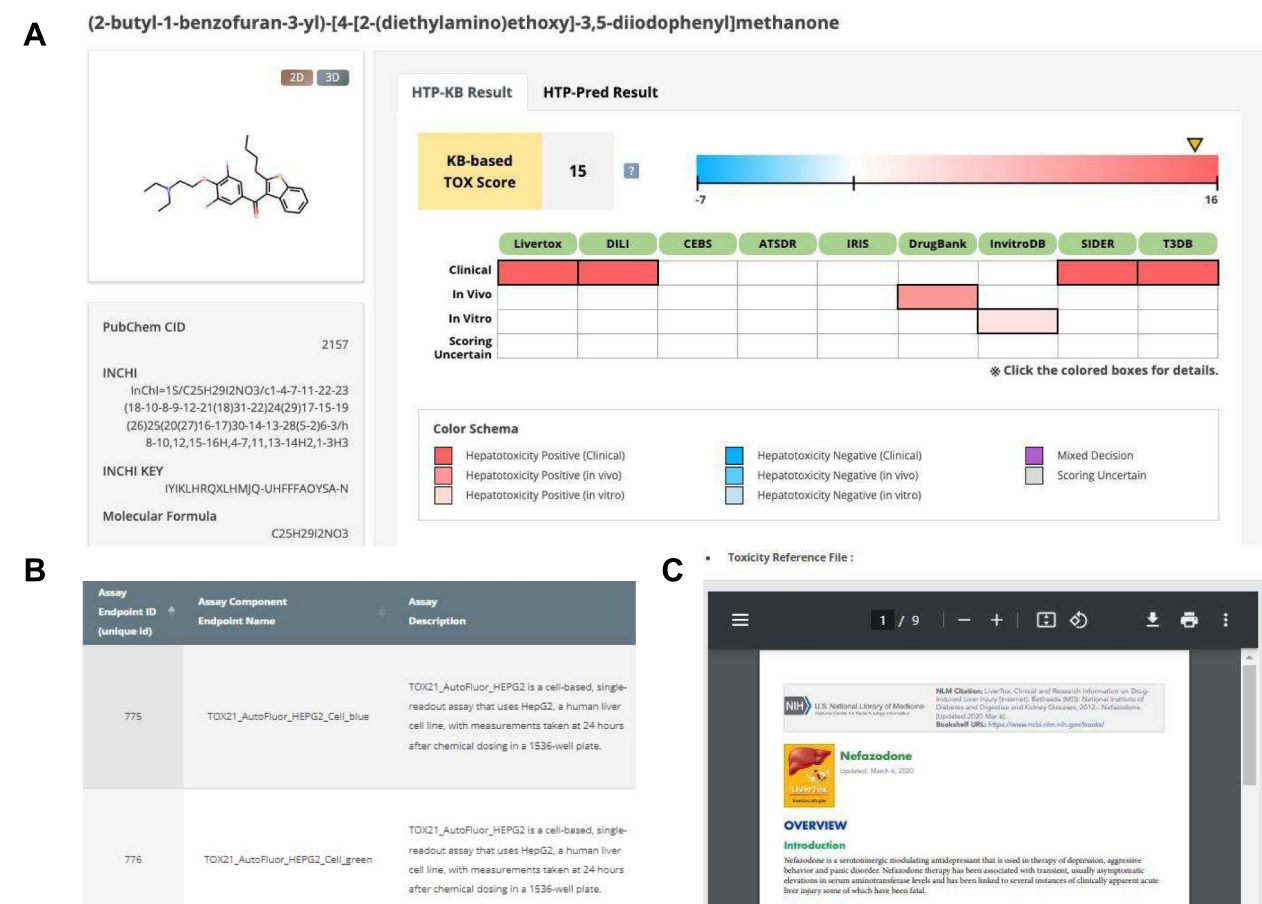


Fig. 5 HTP-KB result of queried compound. **a** Users can expand the compound's property information on the left side. A question mark icon next to the overall score explains score calculation, with a yellow triangle indicating its relative position on the color bar. Colored blocks in the main database table denote available curation sources. Selection of a database highlights it in yellow, revealing detailed information in distinct formats on subpages. **b** InvitroDB in CSV format, and **c** LiverTox in PDF format

HTP-KB化合物的预测评分。为了帮助评估预测结果的可信度, HTP-KB化合物根据整体整理评分被分为三类肝毒性: 阴性(− 7 到0)、中度阳性(0到7)和高度阳性(7到16)。这种分类有助于用户根据已建立的肝毒性分类阈值解读预测评分。

在页面的左侧, 显示了化合物结构, 每个原子的关键性分数通过等高线表示。图底部的详细表格列出了每个原子的关键性分数, 并突出显示了预测肝毒性分数的主要原子。子页面的下半部分展示了毒性基团的结果, 右侧附有详细表格。该表格列出了以SMARTS格式识别出的毒性基团模式, 包括SMARTS模式的来源、数值标识符以及基于原子重要性得出的总分。

分数。多个毒性基团可能对应相同的SMARTS模式, 每个都用一个不同的数字标识符标识。用户可以通过点击表格中的相应行, 方便地验证化合物上每个模式的具体位置。

讨论与结论

肝毒性门户(HTP)代表了在整合全面肝毒性数据和利用最先进技术推进预测建模方面的开创性努力。知识库(HTP-KB)和预测模块(HTP-Pred)旨在解决理解和预测药物诱导肝损伤方面的关键空白。HTP-KB因其丰富的内容和专家策划而突出, 将证据分类为临床、体内和体外类别。独特的肝毒性评分系统将来自多个来源的数据汇总为统一的指标, 提供

prediction scores for HTP-KB compounds. To aid in assessing the confidence of the prediction result, HTP-KB compounds are categorized into three hepatotoxicity classes based on overall curation scores: negative(− 7 to 0), moderately positive(0 to 7), and highly positive(7 to 16). This categorization assists users in interpreting the prediction score relative to established thresholds for hepatotoxicity classification.

On the left side of the page, the compound structure is displayed, with each atom's importance score depicted in contours. A detailed table at the bottom of the figure specifies the importance score for each atom, highlighting the primary atom responsible for predicting the hepatotoxicity score. The lower part of the subpage presents the toxicophores result, accompanied by a detailed table on the right side. This table outlines the identified patterns of the toxicophore in SMARTS format, including the origin of SMARTS patterns, numerical identifiers, and a summation score derived from atom importance

scores. Multiple toxicophores may correspond to the same SMARTS pattern, each identified with a distinct numerical identifier. Users can conveniently verify the location of each pattern highlighted on the compound by clicking the respective rows in the table.

Discussion and conclusions

The HepatoToxicity Portal(HTP) represents a pioneering effort in consolidating comprehensive hepatotoxicity data and advancing predictive modeling using state-of-the-art techniques. Both the knowledgebase(HTP-KB) and prediction modules(HTP-Pred) are designed to address critical gaps in understanding and predicting drug-induced liver injury. HTP-KB stands out for its extensive content and expert curation, classifying evidence into clinical, in vivo, and in vitro categories. A unique hepatotoxicity scoring system aggregates data from multiple sources into a unified metric, providing

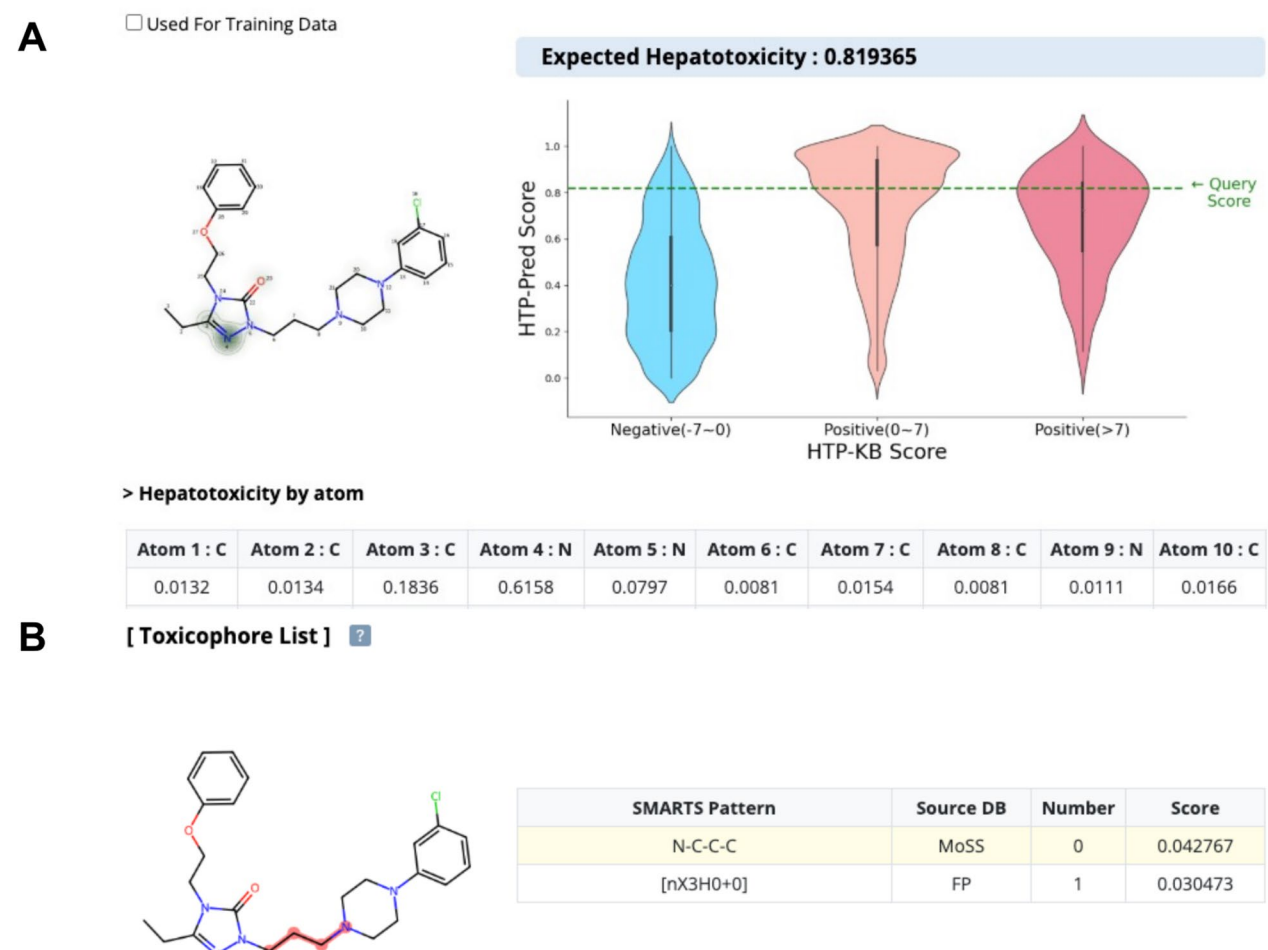
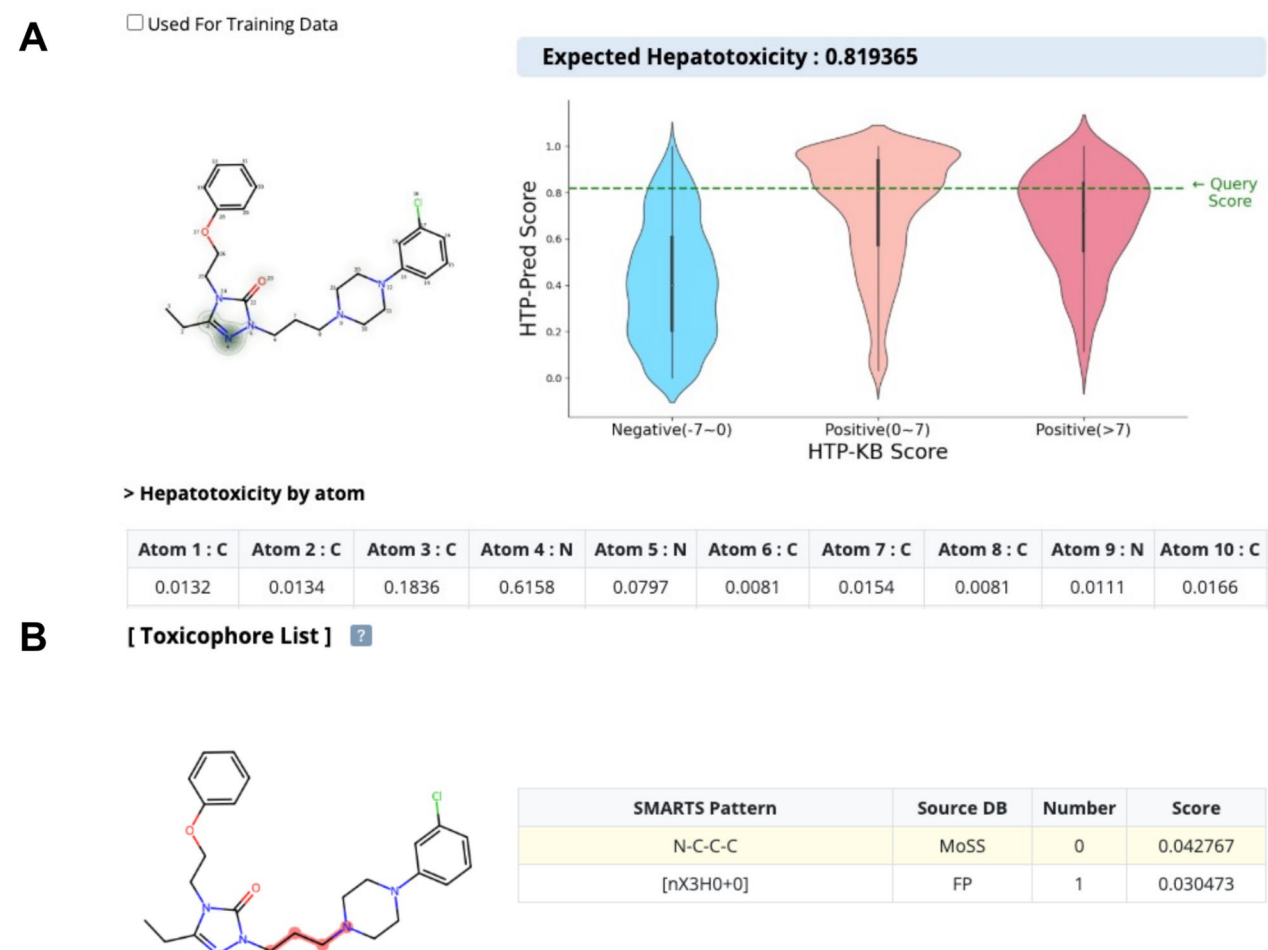


Fig. 6 Result page of HTP-Pred. The HTP-Pred result pages illustrate the predicted toxicity score and the contribution of each atom to toxicity assessment. **a** The HTP-Pred score is displayed with the distribution plot in the upper section. Additionally, the atomic importance scores are visually represented on the compound structure plot. **b** The toxicophore list is accessible through the table with visual representation on the compound structure. Columns include the SMARTS pattern, its source database, and the overall importance score of the substructure. The 'Number' column enumerates instances where multiple substructures correspond to a single SMARTS pattern. Users can interactively highlight specific substructures on the compound plot by selecting corresponding rows in the table

为跨学科的研究人员提供肝毒性化合物的全面概述。

HTP-Pred 利用预训练的 GIN 模型 Mol-CLR, 该模型在来自 PubChem 的约 1000 万条未标记分子数据上进行训练, 并在经过整理的 HTP-KB 数据集上进行了微调。比较评估表明, 与基于传统机器学习的基线模型和其他用于肝毒性预测的网页服务器相比, HTP-Pred 的性能更优。此外, HTP-Pred 支持毒性基团的识别, 使研究人员能够定位具体的分子特征, 这些特征对肝毒性预测有贡献, 从而有助于在药物设计和优化中做出明智决策。然而, 该模型可能会面临源自合并数据库和模型本身的内在偏差。

量化并区分偶然性不确定性和认知性不确定性将为肝毒性预测结果提供更深入的见解。

HTP 网页界面提供对精选数据和预测模型的直观访问, 便于用户顺畅导航, 查找关于肝毒性化合物的详细信息。它包括强大的搜索功能, 并提供来自 HTP-KB 的全面精选信息以及来自 HTP-Pred 的预测结果。

展望未来, HTP 的持续更新和改进有望提升预测能力并扩展数据库覆盖, 满足毒理学和药理学中不断发展的研究需求。HTP 有望通过……对药物研究产生持久影响

researchers across disciplines with a comprehensive overview of hepatotoxic compounds.

HTP-Pred leverages the pre-trained GIN model, Mol-CLR, which is trained on approximately 10 million unlabeled molecular data from PubChem and fine-tuned on the curated HTP-KB dataset. Comparative evaluation demonstrates superior performance compared to traditional ML-based baselines and other web servers for hepatotoxicity prediction. Additionally, HTP-Pred supports the identification of toxicophores, enabling researchers to pinpoint specific molecular features contributing to hepatotoxicity predictions, thereby aiding informed decision-making in drug design and optimization. However, the model may face intrinsic biases arising from the merged databases and the model itself.

Quantifying and distinguishing aleatoric and epistemic uncertainties would provide deeper insights into the hepatotoxicity prediction results.

The HTP web interface provides intuitive access to curated data and predictive models, facilitating seamless navigation for users seeking detailed compound information on hepatotoxicity. It includes robust search functionalities and offers comprehensive curated information from HTP-KB along with prediction results from HTP-Pred.

Looking forward, ongoing updates and enhancements to HTP promise to refine predictive capabilities and expand database coverage, meeting evolving research needs in toxicology and pharmacology. HTP is poised to make a lasting impact on pharmaceutical research by

提供对肝脏毒性机制的关键见解, 并促进更安全、更有效治疗药物的开发。

总之, HTP代表了毒理学和药物开发的重大进展。通过整合来自多个数据库的精心策划的数据并采用最先进的预测模型, HTP为评估与化学化合物相关的肝毒性风险提供了一个全面的资源。它将复杂的数据策划与先进的深度学习方法相结合的能力, 突显了其在提高药物安全性评估和加速治疗创新方面的潜力。总而言之, HTP展示了整合策划数据与先进计算技术的变革性潜力, 为增强药物安全性评估和生物医学研究铺平了道路。

缩写

ADME 吸收、分布、代谢和排泄 AUROC 接收器操作特征曲线
下面积 CASID 化学文摘服务标识符 CID 化合物编号 DB 数据库
DILI 药物性肝损伤 EPA 美国环境保护署 GCN 图卷积网络
GIN 图同构网络 GNN 图神经网络 HTP 肝毒性门户 INCHI 国际化学标识符 IUPAC 国际纯粹与应用化学联合会 KB 知识库
MedDRA 药品监管活动医学词典 ML 机器学习 Pred 预测
RBF 径向基函数 RF 随机森林 SMARTS 任意目标化学结构规范
SMILES 简化分子输入线性表示系统 SSL 自监督学习 SVM 支持向量机

补充信息

在线版本包含可在 <https://doi.org/10.1186/s13321-025-00992-8> 获取的补充材料。

附加文件 1。

致谢

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作者贡献

JH – 数据整理与验证, 撰写—原创草稿, 数据可视化; WZ 和 JL – 软件开发, 撰写—原创草稿; IJ – 网站服务器开发; MJK 和 TDL – 数据整理; SJK 和 KBK – 数据整理, 项目管理; DH – 项目管理, 资金获得; BL – 监督, 网站开发; HSK – 监督, 项目管理, 资金获得; WYK – 监督, 项目管理, 撰写—审阅与编辑; SL – 概念构思, 监督, 项目管理,

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数据和材料的可用性

在本研究过程中未生成或分析任何数据集 udy.

声明

利益冲突

作者声明没有竞争利益。

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providing critical insights into liver toxicity mechanisms and facilitating the development of safer and more effective therapeutic agents.

In conclusion, HTP represents a significant advancement in toxicology and drug development. By integrating curated data from multiple databases and employing cutting-edge predictive models, HTP offers a comprehensive resource for assessing hepatotoxicity risks associated with chemical compounds. Its ability to merge sophisticated data curation with advanced deep learning methodologies underscores its potential to enhance drug safety evaluation and accelerate therapeutic innovation. In summary, HTP exemplifies the transformative potential of integrating curated data and advanced computational techniques, paving the way for enhanced drug safety assessment and biomedical research.

Abbreviations

ADME	Absorption, distribution, metabolism, and excretion
AUROC	Area under the receiver operating characteristic
CASID	Chemical abstracts service identifier
CID	Compound ID
DB	Database
DILI	Drug-induced liver toxicity
EPA	U.S. Environmental Protection Agency
GCN	Graph Convolutional Network
GIN	Graph Isomorphism Network
GNN	Graph Neural Network
HTP	HepatoToxicity Portal
INCHI	International Chemical Identifier
IUPAC	International Union of Pure and Applied Chemistry
KB	Knowledgebase
MedDRA	Medical Dictionary for Regulatory Activities
ML	Machine learning
Pred	Prediction
RBF	Radial basis function
RF	Random forest
SMARTS	SMiles ARbitrary Target Specification
SMILES	Simplified molecular-input line-entry system
SSL	Self-supervised learning
SVM	Support vector machine

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13321-025-00992-8>.

Additional file 1.

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Author contributions

JH – Data curation and validation, Writing—Original draft, Visualization; WZ and JL –Software development, Writing—Original draft; IJ – Webserver development; MJK and TDL – Data curation; SJK and KBK – Data curation, Project administration; DH – Project administration, Funding acquisition; BL – Supervision, Web development; HSK – Supervision, Project management, Funding acquisition; WYK – Supervision, Project administration, Writing—Review & editing; SL – Conceptualization, Supervision, Project administration,

Writing—Review & editing; All authors corrected and approved the final manuscript.

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Availability of data and materials

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

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